D · Teacher guide, alignment and marking

The worksheet group receives no prelesson teaching and does not operate the VL. Topic notes are available from the start. Preserve first responses before distributing the complete answer copy. Correct in another colour without erasing originals. D is for teachers.

Suggested 15–20 minutes; pilot confirmation required

15 minutes: concepts 3, variables/fair comparisons 4, shading/tip removal 3, agar（瓊脂） 3, integration 2. For 20 minutes: 4, 5, 4, 4, 3 respectively. Arrange answer checking after preserving first responses under the shared study procedure; if included in the same period, pilot the allocation without removing core inquiries.

Clarify procedures, vocabulary and how to read figures without revealing answers before first responses. Record additional support and actual time; completion by every learner is not guaranteed. Preserve first and corrected responses with study IDs under the established privacy procedure.

VL sources, concepts and points

Paper labels are A tip shaded, B below-tip shaded, C intact uncovered and D cut tip. Part 1 shows Designs A/B; Part 2 keeps these labels and adds control C last. Alignment: WS A to VL B, WS B to VL C, WS C to VL A, WS D to VL D. Q5 compares C/A, Q6 C/B and Q9 C/D. Q1–14 and points are unchanged.

Q1 variables: #variableChoices, IV 1, DV 1, CV 2 (4 total). Award each category only for the complete correct set. Q2 assumptions: #assumptions and #assumptionDefinition, 1 for all three correct items without the distractor. Q3 control: #controlPlan, 1 for adding an intact, uncovered coleoptile（胚芽鞘） as baseline with other conditions constant.

Q4 shading: #bench / #observationTable, paper A no clear bending, B left, C left: 1 proportionally across three records. Q5–7 #qShade, #qBelow and #qTip: 1 each. Q8 tip removal: #tipBench / #tipObservationTable, paper C yes, D no: 1 proportionally; Q9 #qCap: tick one, 1 point.

Q10 agar（瓊脂）: #extensionBench / #extensionTable. E 0° straight, F 0° straight, G 35° right, H 35° left. Angles 0.5 each (2 total); directions 0.25 each (1 total). E/F require 0°; G/H allow ±2°. Q11–12 #extEF and #extPosition: tick one, 1 each; Q13 #extXAmount, #extCellElongation and #extBendDirection: 1/3 each (1 total).

Q14 integration: #reflection, same diagram as VL; only learning points becomes topic notes, 5 points. Worksheet formative total: 23; it cannot be converted directly to the VL activity total of 29. Use the separately established common assessment for outcomes.

Reference answers and acceptable wording

Variables: IV shading; DV coleoptile（胚芽鞘） bending; CV light direction/intensity, species/initial height/stage, duration, temperature and water. Assumptions: similar initial condition, unrestricted growth and no marked temperature difference; reject species alone removes size control. Control: add an intact, uncovered coleoptile（胚芽鞘） under the same other conditions.

Shading answers in order: no clear bending towards light; can still bend towards light; tip. Tip removal: presence of the tip also affects growth/elongation（延長）. Agar（瓊脂）: the tip may produce a transferable growth-promoting effect; placement affects bending direction; more, more, right. See the complete answer worksheet for every question, option, figure and answer in its original position.

Q9, Q11 and Q12 are single-choice tick questions. Award credit only for selecting the one correct option. Multiple ticks are assessed incorrect; missing responses remain unavailable/pending.

Integrated explanation: one point per causal link

1. The tip senses light from the right.
2. The left side is shaded and has more auxin（生長素）.
3. The growth-promoting effect passes down from the tip.
4. Consequently, cells on the left elongate（延長） more than those on the right.
5. Unequal elongation（延長） causes bending towards the light on the right.

One point per correct causal link, total 5. Accept scientifically sound equivalent wording; do not count keywords or award the same link twice. Missing links receive no point; assessed unrelated responses score 0; unassessed responses remain blank.

Scoring, versions and research limits

VL19 / rubric vl19-causal-5: SPS 24 plus integrated explanation 5 = 29. Inference uses the three main, one tip and three agar（瓊脂） comparison items proportionally (4). Communication assesses clarity of angle/direction records (0–4). Old evidence, reflection-expression and four knowledge fields are not applicable, preventing duplicate causal-content marks. Observation, classification, design and practical work retain their original evidence, 4 each.

Communication: 4 complete and clear; 3 mostly clear; 2 partly clear; 1 little relevant expression; 0 assessed without relevant expression. Assess original predictions for testability and reasoning, not agreement with results. Time, clicks, retries and completion rate do not directly score ability. Paper omits initial observations, prior original predictions, apparatus design and virtual operation: mark not applicable, never substitute zero.

Excel blank means pending, 0 assessed zero, and not applicable does not block totals. Applicable manual columns enforce integer ranges; formulas also reject pasted negatives, excessive values, fractions and text. Totals require submitted reflection and valid applicable scores. Older rows retain their questions, models, graphs, total 32 and rubric; mixed exports use per-row versions. Preserve IDs, original text, snapshots, images, events and queues. Manual marks remain in saved Excel copies. Re-export old Excel files.

Scientific limits and teaching objectives

A coleoptile（胚芽鞘） is part of a grass seedling’s shoot and protects young leaves. Shoots usually show positive phototropism（向光性）. Photoreceptors in tip tissue sense light; auxin（生長素） itself does not detect the light source. In the left-lit mechanism figure, the shaded right side elongates（延長） more and the shoot bends left. The window figure is lit from the right, reversing these sides.

Shading supports tip involvement in sensing light. Cutting also wounds and removes other tissues, so cannot alone establish the sensing mechanism. No elongation（延長） in D is a model setting, not a universal rule for cut plants. E/F supports a transferable growth-promoting effect without identifying auxin（生長素） or its distribution under unilateral light. In darkness, left-placed treated agar（瓊脂） promotes left-side elongation（延長） and bending right, with the reverse for right placement. Central treated agar（瓊脂） produces straight elongation（延長）; central blank agar（瓊脂） gives no clear elongation（延長）.

Objectives: identify variables and controls, read qualitative outcomes and angles, infer sensing and growth effects from the three comparisons, and explain bending causally. Static reading is not dynamic operation; model values are not measurements. Do not restore deleted graphs, prescribed comparisons or extra follow-ups. Check derived answers against the student's own valid readings without cascading penalties; this WS adds no derived question.

This is a draft activity-feedback rubric, without claims of cross-VL equivalence, official syllabus endorsement or validity evidence. Follow the established common assessment and missing-data protocol: missing responses remain unavailable/pending, unasked items are not applicable, assessed incorrect responses score 0. Pilot bilingual reading load, physical printing, marking consistency and the 15–20-minute allocation.